

ARNAV SINGH

+91 7428775691 | arnavwork08@gmail.com | [LinkedIn](#) | [GitHub](#) | [Kaggle](#)

SUMMARY

Final-year Computer Engineering student (CGPA: 8.45) with hands-on software development experience building and deploying end-to-end applications — designing REST APIs, containerizing services with Docker, and shipping them to production on Render and AWS. Comfortable across the full development lifecycle: version control (Git/GitHub), backend API design (FastAPI), SQL-based data layers, and cloud deployment. Currently expanding into modern full-stack (React, Node.js/Express, MongoDB) to complement existing backend and data engineering experience.

EDUCATION

B.Tech, Computer Engineering — CGPA: 8.45

2023 – 2027

ABES Engineering College, AKTU

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Probability & Statistics, Machine Learning

Class XII: 81% | Class X: 92.4% — St. Thomas School, Indirapuram (CBSE)

TECHNICAL SKILLS

Web & Backend: Python, Java, SQL, FastAPI, REST API Design & Deployment, Git/GitHub

Cloud & DevOps: Docker, AWS (SageMaker basics), Render, MLflow, CI-style deployment workflows

Currently Building: HTML, CSS, JavaScript, React.js, Node.js, Express.js, MongoDB (full-stack project in progress — see Projects)

Data & ML: Scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Power BI, EDA, A/B Testing

PROJECTS

Credit Card Fraud Detection — Live REST API | *Python, FastAPI, Docker, Render, MLflow* ([Live Link](#))

- Built and deployed a production REST API on Render with a real-time inference endpoint, containerized with Docker for reproducible deployment.
- Engineered the end-to-end pipeline (SMOTE + XGBoost), reaching 0.9752 AUC-PR and 87.5% recall, saving an estimated £5,338 vs. a no-detection baseline.
- Used SHAP for interpretability and documented findings in business-readable form — practice in translating technical output for non-technical stakeholders.

Legal Query Classifier — Containerized Backend Service | *BERT, Docker, FastAPI* ([Project Link](#))

- Fine-tuned BERT for multi-task legal text classification, achieving 90%+ accuracy while addressing class imbalance with weighted loss.
- Containerized the service with Docker and served it via FastAPI, mirroring a typical backend microservice deployment pattern.
- Evaluated with precision/recall/F1 per class and documented failure modes for future maintainers.

ShipTrack — Full-Stack Logistics Tracking Platform | *React, Node.js, Express.js, MongoDB, Render, Vercel, Git* ([Live Link](#))

- Built a role-based MERN application (React, Node.js, Express, MongoDB) with JWT authentication, a 9-endpoint REST API enforcing data isolation at the query layer, and a live shipment-status timeline, deployed on Render and Vercel.

Customer Churn & Cohort Retention Analysis | *SQL, Python, Power BI, statsmodels* ([Project Link](#))

- Analyzed 540K+ transactions using cohort analysis, CTEs, and window functions to identify early-stage churn as the primary revenue risk.
- Ran hypothesis tests (chi-square, t-test) to validate churn signal significance; findings informed retention strategy.

ACHIEVEMENTS

- Solved 300+ DSA problems across [LeetCode](#) and [CodeChef](#); CodeChef Rating 1614 (3★).
- ECell Eureka Competition — 2nd place at Zonals (IIT Bombay ECell).
- 3rd Place – Retina AI Hackathon 2026.
- 2025 Google Solution Challenge Participant Badge; Google 5-Day Agentic AI Vibe Coding Contributor Badge.

LEADERSHIP & ACTIVITIES

Growth & Content Strategist — Aasira: STEM Beyond Labels

Jul 2024 – Feb 2026

- Grew audience engagement by 35% through targeted campaigns; tracked performance via weekly dashboards in Google Sheets.

Volunteer — GDGoC Member

Oct 2024 – Sep 2025

- Drove 40% increase in event attendance and 30% improvement in content engagement through data-driven campaign optimisation.